Limits of Rational Functions

We saw last time that if $R(x) = \frac{p(x)}{q(x)}$ is a rational function and $a \in \mathbb{R}$ is in the domain of R, then

$$\lim_{x \to a} R(x) = R(a);$$

that is,

$$\lim_{x \to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}.$$

What if a is a point which is not in the domain of R? We know that, for a rational function $R(x) = \frac{p(x)}{q(x)}$, there are at most $\deg q$ points not in the domain, each corresponding to either a hole or vertical asymptote. Of course, if there is a vertical asymptote, then the limit does not exist. However, if there is a hole at a, then $\lim_{x \to a} R(x)$ exists, even though $R(a)$ does not. Since $R(a)$ does not exist, $\lim_{x \to a} R(x) \neq R(a)$. How, then, do we compute $\lim_{x \to a} R(x)$ when R has a hole at a?

Example. Let $f(x) = \frac{x^2-1}{x-1}$ and consider $\lim_{x \to 1} \frac{x^2-1}{x-1}$. Since $x = 1$ is not in the domain of f, we cannot apply Limit Law (iv) to calculate the limit. The function

$$\begin{aligned} f(x) &= \frac{x^2-1}{x-1} \\ &= \frac{(x-1)(x+1)}{x-1} \end{aligned}$$

and the function $g(x) = x+1$ are identical for all values of $x \neq 1$. The graphs of these two functions are shown below

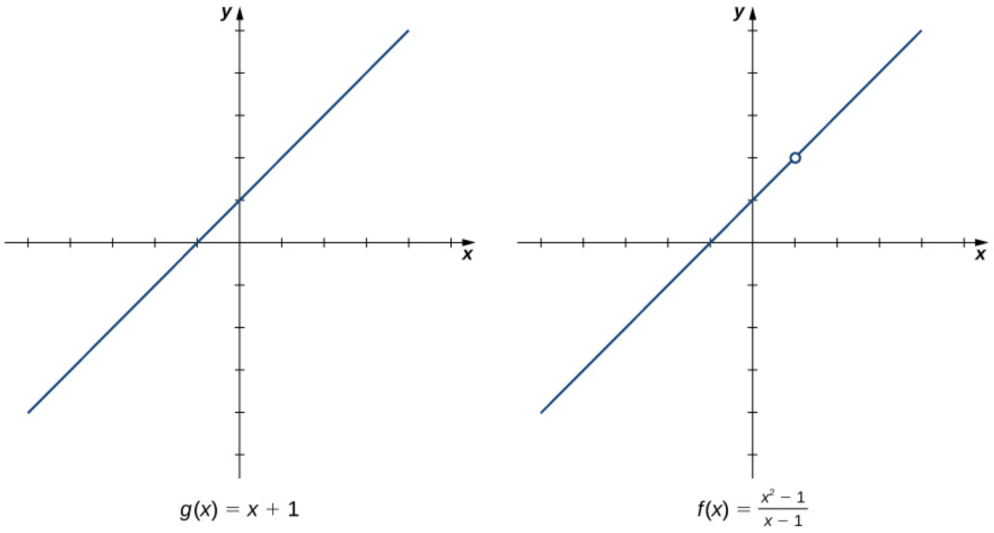


Figure 1: The graphs of $f(x)$ and $g(x)$ are identical for all $x \neq 1$.

Intuitively, f and g should have the same limit as $x \to 1$, since their graphs near (but not at) $x = 1$ both approach the same value. We now prove this is the case. The following theorem says that we

can change the value of a function f at the point a without changing $\lim_{x\to a} f(x)$. Note that the following theorem applies to any functions, not just rational functions.

Theorem. Let f and g be two functions defined on an open interval containing the point a, but perhaps not at a itself. If $f(x) = g(x)$ for all points in this interval whenever $x \neq a$, then $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$.

Proof. Suppose $\lim_{x\to a} f(x) = L$. Then, given $\varepsilon > 0$, there exists $\delta > 0$ such that $0 < |x-a| < \delta$ implies $|f(x) - L| < \varepsilon$. By assumption, $f(x) = g(x)$ for all points x such that $0 < |x-a| < \delta$, so whenever $0 < |x-a| < \delta$, we have $|f(x) - L| = |g(x) - L| < \varepsilon$. This proves that $\lim_{x\to a} g(x) = L = \lim_{x\to a} f(x)$. □

Returning to our example, since the functions $f(x) = \frac{(x-1)(x+1)}{x-1}$ and $g(x) = x+1$ agree at every point except for $x = 1$, by the previous theorem

$$\lim_{x\to 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x\to 1}(x+1) = 1+1 = \boxed{2}.$$

Thus, we may evaluate the limit by first "factoring and canceling", since this does not change the value of the limit.

Example. Let $f(x) = \frac{x^2-3x}{2x^2-5x-3}$ and consider $\lim_{x\to 3} \frac{x^2-3x}{2x^2-5x-3}$. Since the denominator is 0 when $x = 3$, we cannot use the Limit Law 4 to evaluate the limit. Factoring gives

$$\frac{x^2-3x}{2x^2-5x-3} = \frac{x(x-3)}{(x-3)(2x+1)}$$

and therefore

$$\begin{aligned}\lim_{x\to 3} \frac{x^2-3x}{2x^2-5x-3} &= \lim_{x\to 3} \frac{x(x-3)}{(x-3)(2x+1)} \\ &= \lim_{x\to 3} \frac{x}{2x+1} \\ &= \frac{3}{2(3)+1} \\ &= \boxed{\frac{3}{7}}.\end{aligned}$$

Example.

$$\begin{aligned}\lim_{x\to -3}\frac{x^2+4x+3}{x^2-9} &= \lim_{x\to -3}\frac{(x+3)(x+1)}{(x+3)(x-3)}\\ &= \lim_{x\to -3}\frac{x+1}{x-3}\\ &= \frac{-3+1}{-3-3}\\ &= \frac{-2}{-6}\\ &= \boxed{\frac{1}{3}}.\end{aligned}$$

We can use the same technique in the case of a complex fraction, but first we must rewrite it in the form of a rational expression. This can be done by multiplying both the numerator and denominator by the least common denominator of the fractions in the numerator (or denominator):
Example.

$$\begin{aligned}\lim_{x\to 1}\frac{\frac{1}{x+1}-\frac{1}{2}}{x-1} &= \lim_{x\to 1}\frac{(\frac{1}{x+1}-\frac{1}{2})}{x-1}\frac{2(x+1)}{2(x+1)}\\ &= \lim_{x\to 1}\frac{2-(x+1)}{2(x-1)(x+1)}\\ &= \lim_{x\to 1}\frac{2-x-1}{2(x-1)(x+1)}\\ &= \lim_{x\to 1}\frac{1-x}{2(x-1)(x+1)}\\ &= \lim_{x\to 1}\frac{-(x-1)}{2(x-1)(x+1)}\\ &= -\lim_{x\to 1}\frac{(x-1)}{2(x-1)(x+1)}\\ &= -\lim_{x\to 1}\frac{1}{2(x+1)}\\ &= -\frac{1}{2(1+1)}\\ &= \boxed{-\frac{1}{4}}.\end{aligned}$$

Limits of more general algebraic functions

We now consider limits of more general algebraic functions given by expressions consisting of finitely many terms involving the algebraic operations of addition, subtraction, multiplication, division, and raising to a rational power.

We first work out the relevant limit law for a rational power of a function. We will need the following lemma.

Lemma. If

(1) $f(x)$ is defined on an open interval containing a, except perhaps at a itself,

(2) $f(x) \geq 0$ on this interval where it is defined, and

(3) $\lim_{x\to a} f(x) = L$,

then $L \geq 0$.

Proof. Suppose $L < 0$. Then, since $\lim_{x\to a} f(x) = L$, corresponding to $-\frac{L}{2} > 0$ there exists $\delta > 0$ such that $0 < |x-a| < \delta$ implies

$$|f(x)-L| < -\frac{L}{2} \iff \frac{L}{2} < f(x)-L < -\frac{L}{2} \iff \frac{3L}{2} < f(x) < \frac{L}{2} < 0,$$

which contradicts the fact that $f(x) \geq 0$ for all x such that $0 < |x-a| < \delta$. Hence, $L \geq 0$. □

Proposition. If $f(x)$ is defined on an open interval containing a, except perhaps at a itself, $f(x) \geq 0$ on this interval where it is defined, and $\lim_{x\to a} f(x) = L$, then

$$\lim_{x\to a} \sqrt{f(x)} = \sqrt{L}.$$

Call this Limit Law (vii).

Proof. Note that the formula above makes sense due to the lemma. To prove it, let $\varepsilon > 0$. Then

$$\begin{aligned} |\sqrt{f(x)} - \sqrt{L}| &= \left| (\sqrt{f(x)} - \sqrt{L}) \frac{(\sqrt{f(x)} + \sqrt{L})}{\sqrt{f(x)} + \sqrt{L}} \right| \\ &= \frac{|f(x)-L|}{\sqrt{f(x)} + \sqrt{L}} \\ &\leq \frac{|f(x)-L|}{\sqrt{L}} \quad (\text{since } \sqrt{f(x)} \geq 0). \end{aligned}$$

Since $\lim_{x\to a} f(x) = L$, given $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that $0 < |x-a| < \delta$ implies $|f(x)-L| < \sqrt{L}\varepsilon$ and therefore $0 < |x-a| < \delta$ implies

$$|\sqrt{f(x)} - \sqrt{L}| \leq \frac{|f(x)-L|}{\sqrt{L}} < \frac{\sqrt{L}\varepsilon}{\sqrt{L}} = \varepsilon.$$

Example. Applying the Limit Laws,

$$\begin{aligned}\lim_{x\to 6}(2x-1)\sqrt{x+4} &= \lim_{x\to 6}(2x-1)\lim_{x\to 6}\sqrt{x+4}\\ &= \lim_{x\to 6}(2x-1)\sqrt{\lim_{x\to 6}(x+4)}\\ &= (2(6)-1)\sqrt{6+4}\\ &= \boxed{11\sqrt{10}}.\end{aligned}$$

Corollary. If n is a positive integer and $\lim_{x\to a} f(x) = L$, then

$$\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{L}.$$

Here, $f(x)$ is required to be nonnegative if n is even, in which case L is also nonnegative.

Proof. Since

$$L = \lim_{x\to a} f(x) = \lim_{x\to a}(\sqrt[n]{f(x)})^n = \left(\lim_{x\to a}\sqrt[n]{f(x)}\right)^n,$$

the result follows by taking the nth root of each side. □

Corollary. If m,n is a positive integer and $\lim_{x\to a} f(x) = L$, then

$$\lim_{x\to a}(f(x))^{\frac{m}{n}} = L^{\frac{m}{n}}.$$

Here, $f(x)$ is required to be nonnegative if n is even, in which case L is also nonnegative.

Proof. By the previous Corollary, $L^{\frac{1}{n}} = \lim_{x\to a} f(x)^{\frac{1}{n}}$. Raising both sides to the mth power and using the Product Law gives

$$L^{\frac{m}{n}} = \left(\lim_{x\to a} f(x)^{\frac{1}{n}}\right)^m = \lim_{x\to a}(f(x)^{\frac{1}{n}})^m = \lim_{x\to a} f(x)^{\frac{m}{n}}.$$

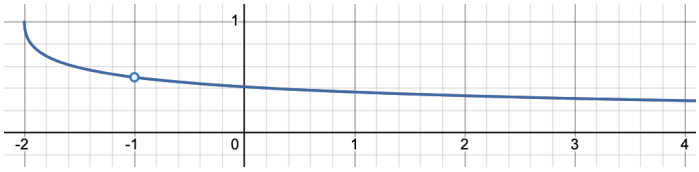
□

Example.

$$\lim_{x\to 1}(x+7)^{\frac{2}{3}} = \left(\lim_{x\to 1}(x+7)\right)^{\frac{2}{3}} = (8)^{\frac{2}{3}} = (2)^2 = \boxed{4}.$$

Limits of algebraic functions at points not in the domain

Example. Consider $\lim_{x\to -1}\frac{\sqrt{x+2}-1}{x+1}$. Since $\lim_{x\to -1}(\sqrt{x+2}-1)=0$ and $\lim_{x\to -1}(x+1)=0$, Limit Law (iv) cannot be used. Looking at the graph, there is a hole at $x=-1$, so the limit exists.



How can we calculate it? In this case, we can try rationalizing:

$$\begin{aligned}\frac{\sqrt{x+2}-1}{x+1} &= \frac{\sqrt{x+2}-1}{x+1}\cdot\frac{\sqrt{x+2}+1}{\sqrt{x+2}+1}\\ &= \frac{x+2-1}{(x+1)(\sqrt{x+2}+1)}\\ &= \frac{x+1}{(x+1)(\sqrt{x+2}+1)}\end{aligned}$$

Since $\frac{\sqrt{x+2}-1}{x+1}$ and $\frac{x+1}{(x+1)(\sqrt{x+2}+1)}$ differ only at $x=-1$ they have the same limit, so we have

$$\begin{aligned}\lim_{x\to -1}\frac{\sqrt{x+2}-1}{x+1} &= \lim_{x\to -1}\frac{x+1}{(x+1)(\sqrt{x+2}+1)}\\ &= \lim_{x\to -1}\frac{1}{\sqrt{x+2}+1}\\ &= \frac{\lim_{x\to -1}1}{\lim_{x\to -1}(\sqrt{x+2}+1)}\\ &= \frac{1}{\sqrt{-1+2}+1}\\ &= \frac{1}{\sqrt{1}+1}\\ &= \boxed{\frac{1}{2}}.\end{aligned}$$

Example.

$$\begin{aligned}
\lim_{x\to 5}\frac{\sqrt{x-1}-2}{x-5} &= \lim_{x\to 5}\frac{\sqrt{x-1}-2}{x-5}\cdot\frac{\sqrt{x-1}+2}{\sqrt{x-1}+2}\\
&= \lim_{x\to 5}\frac{x-1-4}{(x-5)(\sqrt{x-1}+2)}\\
&= \lim_{x\to 5}\frac{x-5}{(x-5)(\sqrt{x-1}+2)}\\
&= \lim_{x\to 5}\frac{1}{\sqrt{x-1}+2}\\
&= \frac{1}{\sqrt{5-1}+2}\\
&= \frac{1}{\sqrt{4}+2}\\
&= \boxed{\frac{1}{4}}.
\end{aligned}$$